

FluidicSwarm: Embodiment of Swarm Robots Using Fluid Behavior Imitation

Michikuni Eguchi
Cluster Metaverse Lab
Shinagawa, Japan
University of Tsukuba
Shinagawa, Japan
m.eguchi@cluster.mu

Mai Nishimura
OMRON SINIC X Corporation
Tokyo, Japan
mai.nishimura@sinicx.com

Shigeo Yoshida
OMRON SINIC X Corporation
Tokyo, Japan
shigeo.yoshida@sinicx.com

Takefumi Hiraki
Cluster Metaverse Lab
Tokyo, Japan
t.hiraki@cluster.mu

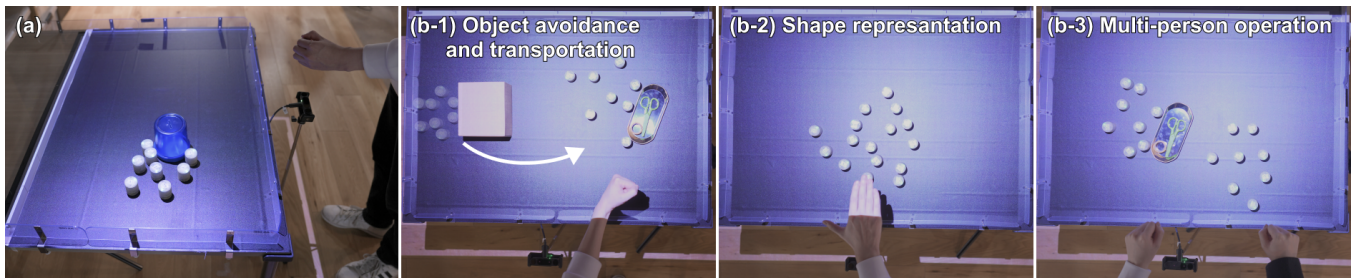


Figure 1: (a) FluidicSwarm achieves an intuitive swarm robot control experience by imitating fluid dynamics and replicating hand shapes with the swarm robots. This technology enables the following applications: (b-1) Flexible switching between object avoidance and transportation by changing the flexibility of the swarm formation. (b-2) Expressing various swarm formations by imitating hand shapes. Using fluid dynamics, the robots' arrangement is determined automatically in real time. (b-3) Collaborative swarm robot control by multiple users.

Abstract

This paper introduces “FluidicSwarm”, a swarm robot control system that enhances the operability of swarm robots by embodying fluid-like swarm robots as an extension of the user’s body. In FluidicSwarm, users can easily manipulate the characteristics of the fluid through hand movements, changing the shape and flexibility of the swarm robot formation. This capability enables the efficient execution of various tasks, such as obstacle avoidance and object transportation, and simplifies the swarm robot arrangement problem in shape representation. Consequently, users can operate swarm robots with greater efficiency in a variety of environments, markedly enhancing the quality of swarm robot control experiences, including teleoperation and remote collaboration.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

swarm robotics, tangible interaction, embodiment

ACM Reference Format:

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1 Introduction

Swarm robotics refers to a collective of numerous robots that possess high task execution capabilities, exceeding those of individual robots through the formation of aggregates. Furthermore, these collectives exhibit great versatility by dispersing and reforming as needed to accomplish various tasks. These characteristics are particularly valuable in applications within the field of human-computer interaction (HCI), such as teleoperation [Ichihashi et al. 2024; Ihara

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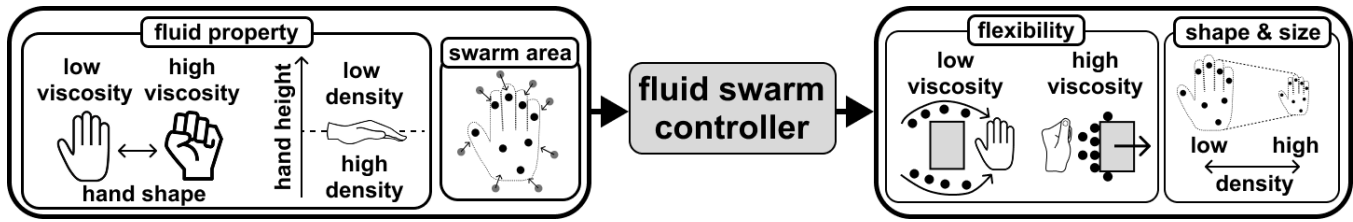


Figure 2: Proposed swarm robot controller design. This controller takes fluid properties and the swarm robot area as inputs. These inputs are determined by the user’s hand shape, height, and area, which are captured by a hand-tracking camera. Based on these inputs, the controller changes the flexibility, shape, and size of the swarm.

et al. 2023] and tangible user interfaces [Le Goc et al. 2016], where the execution of diverse tasks is required.

Swarm robots have the potential to serve as a dynamic, reconfigurable interface for human interaction. However, treating a swarm as an interface presents three fundamental challenges. First, as the number of robots increases, intuitive control becomes increasingly difficult. Second, tasks involving complex shape formation or interaction with objects remain intractable. While conventional gesture-based methods allow users to specify target locations for the swarm [Ihara et al. 2023; Le Goc et al. 2016], these methods are limited to basic movements and simple formations. Finally, more advanced tasks, such as object avoidance and transportation, require intricate movement instructions and close coordination among multiple robots, further complicating the control process.

To address these challenges, we propose “FluidicSwarm,” a swarm robot control system that enhances the operability of swarm robots by embodying fluid-like swarm robots as an extension of the user’s body. In FluidicSwarm, fluid-like swarm behaviors enable automatic control of robot arrangement and movement based on fluid properties and shape instructions, without detailed movement instructions from the user. This allows users to seamlessly perform various tasks such as obstacle avoidance, object transportation, and complex shape formation by simply adjusting the fluid’s properties and shape through hand movements.

2 System Design

Our developed swarm robot control system generates various swarm behaviors in response to human hand movements via fluid dynamics (Fig. 2). This swarm robot controller employs the Smoothed Particle Hydrodynamics (SPH), commonly used in fluid simulators, to achieve swarm control that imitates fluid motion [Eguchi et al. 2024]. The controller receives fluid properties, determined by hand movements, and swarm area data as inputs. First, the fluid property factor manipulates the flexibility and size of the swarm formation by changing the viscosity and density of the fluid that the swarm robot imitates. This capability allows the swarm to efficiently perform various tasks in complex environments by softening the formation during avoidance maneuvers and stiffening it during transport operations. Furthermore, the viscosity and density of the fluid are determined through simple hand operations, such as hand shape and height changes. By combining these intuitive movements, users can flexibly and intuitively generate swarm behaviors that adapt to various tasks. Next, the swarm area defines the formation shape of the swarm. The controller determines the robot arrangement as

if the given area were being filled with fluid. This allows users to achieve highly flexible swarm formations through hand shape representation without considering the detailed positioning of robots.

In this system, we use the two-wheeled swarm robot “maru”¹, which communicates with the host computer via 2.4 GHz wireless. The robot sends its position data to the host computer, then the system updates swarm movements by continuously sending control commands to each robot.

3 Application

Our system will benefit the operability of the swarm robot. We demonstrated three swarm robot applications (Fig. 1):

- **Object Avoidance and Transportation:** Our control system can flexibly change the behavior of the swarm formation in response to objects by modifying the formation’s flexibility.
- **Shape Representation:** Users can express various swarm formation shapes using their hands. Furthermore, by combining multiple hands, users can represent more complex shapes.
- **Multi-Person Operation:** Multiple users can control several swarms simultaneously by moving their hands. This capability enables more complex and adaptive swarm operations.

Acknowledgments

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¹<https://github.com/omron-sinix/maru>